UNIT III : Two-Dimensional Analysis

3.1 Introduction

Two-dimensional problems are solved when the body can be idealised in one plane : **plane stress** (thin plate in its own plane, $\sigma_z = 0$), **plane strain** (long prismatic body, $\varepsilon_z = 0$) or **axisymmetric** solids. Each node has **2 dof** (u, v). The simplest 2-D element is the **3-noded triangle** - the **Constant Strain Triangle (CST)**.

- Triangles fit curved and irregular boundaries easily and are used for automatic meshing.
- Nodes are numbered **anticlockwise** inside each element so that the area A is positive.
- Element dof vector $\{q\} = [q_1\ q_2\ q_3\ q_4\ q_5\ q_6]^T = [u_1\ v_1\ u_2\ v_2\ u_3\ v_3]^T$.

CST: 3 nodes, 6 dof
$\{q\} = [q_1 \dots q_6]^T$
ε = const over element

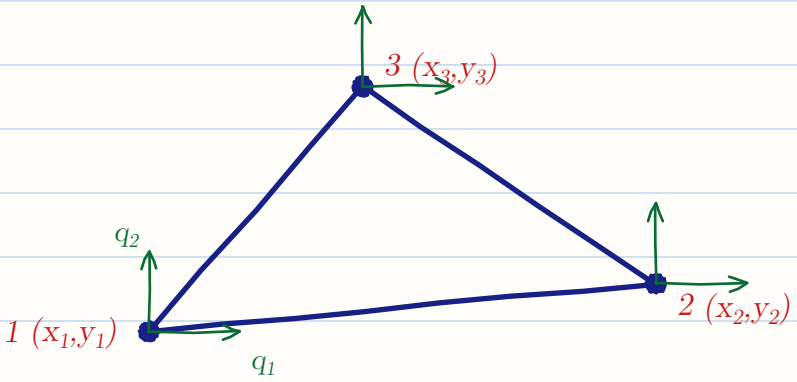


Fig 3.1 Constant strain triangle

3.2 Displacement field and shape functions

Three nodes per variable ⇒ 3 constants, a complete linear polynomial : $u = \alpha_1 + \alpha_2 x + \alpha_3 y$, $v = \alpha_4 + \alpha_5 x + \alpha_6 y$. Solving with nodal values :

$$u = N_1u_1 + N_2u_2 + N_3u_3,\ v = N_1v_1 + N_2v_2 + N_3v_3$$

$$N_i = (a_i + b_i x + c_i y)/2A \quad (i = 1,2,3)$$

with (cyclic order 1→2→3) : $a_1 = x_2y_3 - x_3y_2$, $b_1 = y_2 - y_3$, $c_1 = x_3 - x_2$ (and similarly for 2, 3 by cycling the subscripts), and

$$2A = x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2) = \det [1\ x_1\ y_1\ ;\ 1\ x_2\ y_2\ ;\ 1\ x_3\ y_3]$$

Properties : $N_i = 1$ at node i, 0 at the others ; $N_1+N_2+N_3 = 1$; N_i varies linearly and is identical to the **area co-ordinate** L_i = (area of sub-triangle opposite node i)/A. Along an edge the displacement is linear, hence the displacement is continuous between adjacent elements (compatible).

$$\{u,v\}^T = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 \end{bmatrix}$$

3.3 Strain-displacement matrix and stiffness matrix

$\varepsilon_x = \partial u/\partial x$, $\varepsilon_y = \partial v/\partial y$, $\gamma_{xy} = \partial u/\partial y + \partial v/\partial x$. Since $\partial N_i/\partial x = b_i/2A$ and $\partial N_i/\partial y = c_i/2A$:

$$[B] = 1/(2A) \begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & c_1 & 0 & c_2 & 0 & c_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix}$$

Key point: *[B] contains only nodal coordinates - it is* **constant** *over the element. Hence strains and stresses are constant ⇒ the name CST. The stress is best interpreted at the centroid; fine meshes are needed where stress gradients are high.*

$$\{\varepsilon\} = [B]\{q\} \quad \{\sigma\} = [D][B]\{q\}$$

$$[k] = \int [B]^T[D][B]\, t\, dA = t\, A\, [B]^T[D][B] \; (6 \times 6, \text{symmetric})$$

[D] is the plane-stress or plane-strain matrix of Unit I; t = thickness (for plane strain take t = 1 per unit length).

Load vector

- ***Body force*** *(X, Y per unit volume, e.g. weight $Y = -\rho g$) : $\{f_b\} = (tA/3)\,[X\ Y\ X\ Y\ X\ Y]^T$ (one third of the total force at each node).*
- ***Traction (distributed edge load)*** *T_x, T_y per unit area on edge i-j of length L_{ij} : $\{f_T\} = (t\, L_{ij}/2)\,[T_x\ T_y\ T_x\ T_y]^T$ on the two edge nodes (zero at the third node). For a linearly varying load integrate with N_i.*
- ***Point loads*** *are applied directly at the nodes (always place a node under a concentrated load).*
- ***Initial strain / thermal*** *: $\{f_{th}\} = t\, A\, [B]^T[D]\{\varepsilon_0\}$, $\{\varepsilon_0\} = \alpha\Delta T\,[1\ 1\ 0]^T$ (plane stress).*

Treatment of boundary conditions

Global equation $[K]\{Q\} = \{F\}$ with size $2 \times$ (number of nodes). **Elimination** : delete rows and columns of restrained dof ($u = 0$ or $v = 0$). **Penalty** : add large C to the diagonal term. For an **inclined roller** support, use a multi-point constraint (transformation) or penalty : $\beta_1 Q_i + \beta_2 Q_j = \beta_0$. Reactions : $\{R\} = [K]\{Q\} - \{F\}$ for the restrained rows.

Stress calculation

$$\sigma_{1,2} = (\sigma_x+\sigma_y)/2 \pm \sqrt{[\ ((\sigma_x-\sigma_y)/2)^2 + \tau_{xy}^{\ 2}\ ]},\ \tan 2\varphi = 2\tau_{xy}/(\sigma_x-\sigma_y)$$

$$\text{Von Mises} : \sigma_{vm} = \sqrt{(\ \sigma_x^{\ 2} - \sigma_x\sigma_y + \sigma_y^{\ 2} + 3\tau_{xy}^{\ 2}\ )}\ \text{(plane stress)}$$

3.4 Solved Problem 1 : stresses in one triangle

Triangle 1 (0,0), 2 (40,0), 3 (0,30) mm, $t = 5$ mm, $E = 2\times10^5$ N/mm^2, $\nu = 0.3$, plane stress. Nodal displacements $u_2 = 0.02$ mm, $v_3 = 0.015$ mm, all others zero. Find strains and stresses.

$A = ½\times40\times30 = 600$ mm^2. $b_1 = -30$, $b_2 = 30$, $b_3 = 0$; $c_1 = -40$, $c_2 = 0$, $c_3 = 40$.

$$[B] = 1/1200 \begin{bmatrix} -30 & 0 & 30 & 0 & 0 & 0 \\ 0 & -40 & 0 & 0 & 0 & 40 \\ -40 & -30 & 0 & 30 & 40 & 0 \end{bmatrix}$$

Strains $\{\varepsilon\} = [B]\{q\}$: $\varepsilon_x = b_2u_2/2A = 30\times0.02/1200 = 0.00050$, $\varepsilon_y = c_3v_3/2A = 40\times0.015/1200 = 0.00050$, $\gamma_{xy} = (c_2u_2 + b_3v_3 + \ldots)/2A = 0.00000$.

Stresses $\{\sigma\} = [D]\{\varepsilon\}$ with $[D] = (E/(1-\nu^2))\ [1\ \nu\ 0\ ;\ \nu\ 1\ 0\ ;\ 0\ 0\ (1-\nu)/2]$: $\sigma_x = 142.86$ N/mm^2, $\sigma_y = 142.86$ N/mm^2, $\tau_{xy} = 0.00$ N/mm^2.

Principal stresses $\sigma_1 = 142.86$, $\sigma_2 = 142.86$ N/mm^2; von Mises $= 142.86$ N/mm^2.

3.5 Solved Problem 2 : plate with two CST elements

Rectangular plate 40 × 30 mm, $t = 5$ mm, $E = 2\times10^5$ N/mm^2, $\nu = 0.3$ (plane stress). Left edge (nodes 1, 4) fixed; vertical load $P = 1000$ N downward at node 3. Elements : (1) nodes 1-2-3, (2) nodes 1-3-4 (both anticlockwise).

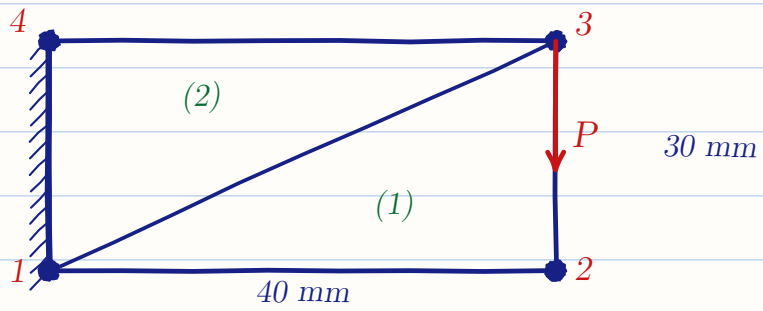


Fig 3.2

Element 1 : (0,0), (40,0), (40,30); ***Element 2*** : (0,0), (40,30), (0,30). Both have area $A = 600$ mm^2.

Element 1 : b = [-30, 30, 0], c = [0, -40, 40]. Element 2 : b = [0, 30, -30], c = [-40, 0, 40].

$[k] = tA[B]^T[D][B]$ for each element is a 6×6 matrix (D for plane stress : $E/(1-\nu^2)$ $= 2.198\times10^5$). Assembling (element 1 dof = 1,2,3,4,5,6 ; element 2 dof = 1,2,5,6,7,8) gives the 8×8 [K]. Boundary conditions $u_1 = v_1 = u_4 = v_4 = 0$ leave Q_3, Q_4, Q_5, Q_6 :

$$10^5 \times \begin{bmatrix} 6.68 & -3.57 & -2.56 & 1.65 \\ -3.57 & 8.77 & 1.92 & -7.33 \\ -2.56 & 1.92 & 6.68 & 0 \\ 1.65 & -7.33 & 0 & 8.77 \end{bmatrix} \begin{Bmatrix} u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = [0,\ 0,\ 0,\ -1000]^T$$

(entries shown in units of 10^5 N/mm). Solution :

u_2	v_2	u_3	v_3
-0.00090 mm	-0.00461 mm	0.00098 mm	-0.00483 mm

Stresses (N/mm^2) from $\{\sigma\} = [D][B]\{q\}$:

Element	σ_x	σ_y	τ_{xy}	σ_1	σ_2	σ_{vm}
1	-5.40	-3.04	-4.05	0.00	-8.44	8.44
2	5.40	1.62	-9.28	12.98	-5.96	16.78

Reactions at the fixed edge : R_{1x} = 1333.3, R_{1y} = 141.8, R_{4x} = -1333.3, R_{4y} = 858.2 N. ΣF_y = 1000.0 N (balances P = 1000 N) - check.

Remark: With only two CST elements the solution is stiff and inaccurate in bending (beam theory gives a much larger tip deflection). Refine the mesh : the CST converges slowly for bending-dominated problems. Stress at the support is not captured; compare with $\sigma = My/I$ at the root.

3.6 Axisymmetric Solids

A solid of revolution loaded symmetrically about its axis (cylinders, pressure vessels, pipes, flywheels, rotors, nozzles, pistons) is analysed on a ***single r-z cross-section***. The geometry, material, loads and BC do not vary with θ, so the displacement has two components : u (radial) and w (axial), $v_\theta = 0$.

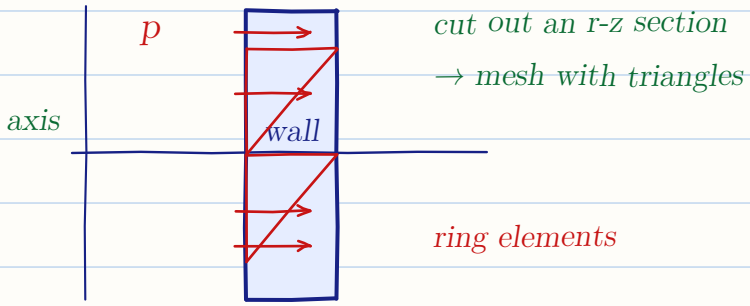


Fig 3.3 Axisymmetric solid → ring elements

The 2-D element is a triangle in the r-z plane, which, when revolved through 360°, becomes a ***ring*** (torus of triangular section) joined to others along nodal circles. Nodal dof : (u_i, w_i) with $\{q\} = [u_1\ w_1\ u_2\ w_2\ u_3\ w_3]^T$.

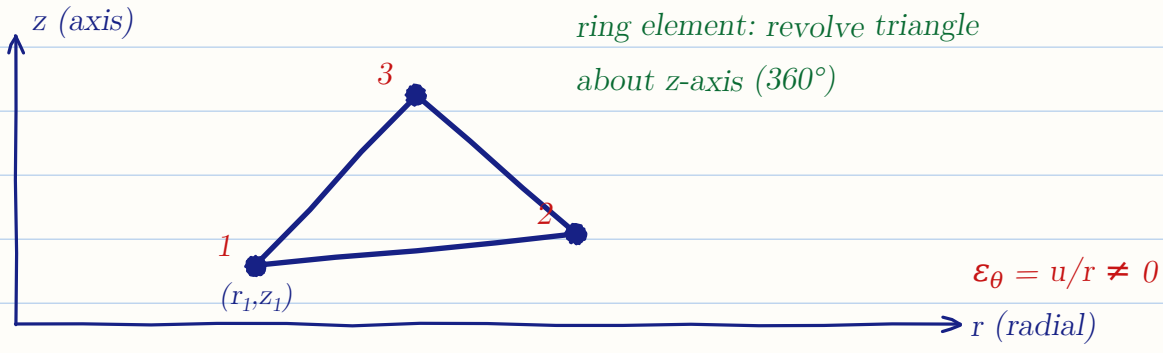


Fig 3.4 Triangular ring element in the r-z plane

Strain-displacement relations

$$\{\varepsilon\} = [\varepsilon_r \; \varepsilon_z \; \varepsilon_\theta \; \gamma_{rz}]^T = \begin{bmatrix} \partial u/\partial r \\ \partial w/\partial z \\ u/r \\ \partial u/\partial z + \partial w/\partial r \end{bmatrix}$$

The hoop strain $\varepsilon_\theta = u/r$ is the new feature : a radial displacement stretches the circumference. Therefore there are **4 stress/strain components** - σ_r, σ_z, σ_θ, τ_{rz}.

Stress-strain matrix

$$[D] = E(1-\nu)/((1+\nu)(1-2\nu)) \begin{bmatrix} 1 & \nu/(1-\nu) & \nu/(1-\nu) & 0 \\ \nu/(1-\nu) & 1 & \nu/(1-\nu) & 0 \\ \nu/(1-\nu) & \nu/(1-\nu) & 1 & 0 \\ 0 & 0 & 0 & (1-2\nu)/(2(1-\nu)) \end{bmatrix}$$

Shape functions and [B] matrix

Same as the CST with (x, y) → (r, z) : $N_i = (\alpha_i + \beta_i r + \gamma_i z)/2A$, $\alpha_1 = r_2 z_3 - r_3 z_2$, $\beta_1 = z_2 - z_3$, $\gamma_1 = r_3 - r_2$ (cyclic).

$$[B] = 1/(2A) \begin{bmatrix} \beta_1 & 0 & \beta_2 & 0 & \beta_3 & 0 \\ 0 & \gamma_1 & 0 & \gamma_2 & 0 & \gamma_3 \\ f_1 & 0 & f_2 & 0 & f_3 & 0 \\ \gamma_1 & \beta_1 & \gamma_2 & \beta_2 & \gamma_3 & \beta_3 \end{bmatrix}$$

where $f_i = \alpha_i/r + \beta_i + \gamma_i z/r$. Because f_i varies with (r, z), [B] is not constant; it is evaluated at the **centroid** $\bar{r} = (r_1+r_2+r_3)/3$, $\bar{z} = (z_1+z_2+z_3)/3$ (good approximation for fine mesh). Hence the element stresses are taken constant and :

$$[k] = 2\pi \, \bar{r} \, A \, [B]^T[D][B] \; (6 \times 6)$$

Remember: Volume = $2\pi \, \bar{r} \, A$ (Pappus theorem). There is no thickness t; the $2\pi \, \bar{r}$ factor replaces it. The element near the axis ($r \to 0$) needs care : the term u/r is singular; use $\bar{r}$ at the centroid, fix $u = 0$ on the axis.

Load vectors

- **Body force** (e.g. centrifugal force $\rho\omega^2 r$ in radial direction, weight along z) : $\{f\} = (2\pi\ \bar{r}\ A\ /3)\ [F_r\ F_z\ F_r\ F_z\ F_r\ F_z]^T$ with $F_r = \rho\omega^2\bar{r}$.
- **Surface pressure** p on an edge i-j of length L_e (e.g. internal pressure on the bore) : $\{f_i\} = (\pi\ p\ L_e/3)(2r_i + r_j)$, $\{f_j\} = (\pi\ p\ L_e/3)(r_i + 2r_j)$, applied in the direction normal to the surface (radial for the bore). Simplified : $f = 2\pi\ \bar{r}_{edge}\ p\ L_e/2$ at each node.
- **Ring / line loads** : a load P per unit circumference at radius r gives a nodal force $2\pi\ r\ P$.

3.7 Solved Problem 3 : axisymmetric triangle

Triangle with nodes 1 (r=20, z=0), 2 (30, 0), 3 (20, 10) mm. $E = 2\times10^5$ N/mm^2, $\nu = 0.3$. Find [B], [D] and [k].

$A = ½\times10\times10 = 50$ mm^2; $\bar{r} = (20+30+20)/3 = 23.33$ mm, $\bar{z} = 3.33$ mm. α = [300, -200, 0], β = [-10, 10, 0], γ = [-10, 0, 10].

$[D] = 2.692\times10^5 \times$ [1 0.4286 0.4286 0 ; 0.4286 1 0.4286 0 ; 0.4286 0.4286 1 0 ; 0 0 0 0.2857] , with $E(1-\nu)/((1+\nu)(1-2\nu)) = 2.692\times10^5$ N/mm^2.

[B] (evaluated at the centroid), entries $\times\ 10^{-2}$:

$$[B] = \begin{bmatrix} -10.000 & 0.000 & 10.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & -10.000 & 0.000 & 0.000 & 0.000 & 10.000 \\ 1.429 & 0.000 & 1.429 & 0.000 & 1.429 & 0.000 \\ -10.000 & -10.000 & 0.000 & 10.000 & 10.000 & 0.000 \end{bmatrix}$$

$[k] = 2\pi\ \bar{r}\ A\ [B]^T[D][B]$ with $2\pi\ \bar{r}\ A = 2\pi\times23.33\times50 = 7330$ mm^3. Result in units of 10^6 N/mm :

$$[k] = \begin{bmatrix} 23.361 & 12.889 & -19.333 & -5.639 & -6.444 & -7.250 \\ 12.889 & 25.374 & -9.666 & -5.639 & -6.847 & -19.736 \\ -19.333 & -9.666 & 22.555 & 0.000 & 1.611 & 9.666 \\ -5.639 & -5.639 & 0.000 & 5.639 & 5.639 & 0.000 \\ -6.444 & -6.847 & 1.611 & 5.639 & 6.042 & 1.208 \\ -7.250 & -19.736 & 9.666 & 0.000 & 1.208 & 19.736 \end{bmatrix}$$

Check : symmetric and singular - the sum of the radial-dof rows is not zero because rigid-body radial translation is not a rigid-body mode of a ring (it stretches the hoop).

Comparison : plane stress / plane strain / axisymmetric

Item	Plane stress / strain	Axisymmetric
Coordinates	x, y	r, z
dof / node	u, v	u, w
Strains	3	4 (adds $\varepsilon_\theta = u/r$)
[B] matrix	constant	varies - use centroid
[k]	$t\ A\ B^T D B$	$2\pi\ \bar{r}\ A\ B^T D B$
Typical use	plates, dams	vessels, rotors, pipes

3.8 More on CST and Axisymmetric Elements

Derivation of the CST shape functions

1. Write $u = \alpha_1 + \alpha_2 x + \alpha_3 y$ and apply $u = u_1, u_2, u_3$ at the three nodes : $\{u_1\ u_2\ u_3\}^T = [\ 1\ x_1\ y_1\ ;\ 1\ x_2\ y_2\ ;\ 1\ x_3\ y_3\ ]\{\alpha_1\ \alpha_2\ \alpha_3\}^T$.
2. Invert the 3×3 matrix; its determinant equals $2A$. The inverse $= (1/2A)\ [\ a_1\ a_2\ a_3\ ;\ b_1\ b_2\ b_3\ ;\ c_1\ c_2\ c_3\ ]$ (co-factor matrix).
3. Substitute $\{\alpha\}$ back : $u = [1\ x\ y](1/2A)[...]\{u_i\} \Rightarrow N_i = (a_i + b_i x + c_i y)/2A$. Same for v.
4. Check : at node 1, $N_1 = (a_1 + b_1 x_1 + c_1 y_1)/2A = 2A/2A = 1$.

Integration with area co-ordinates

$$\int_A L_1^{\,a}\, L_2^{\,b}\, L_3^{\,c}\, dA = a!\ b!\ c!\ 2A\ /\ (a + b + c + 2)!$$

Examples : $\int L_i\, dA = A/3$ (body force $\Rightarrow$ one-third at each node) ; along an edge $\int L_i\, ds = L_e/2$; $\int L_i L_j\, dA = A/12$ $(i \neq j)$, $\int L_i^2\, dA = A/6$. Used for load vectors and mass matrices.

Problem 4 : load vectors for a triangle

Triangle (0,0), (40,0), (0,30), $t = 5$ mm. (a) **Self weight** : steel $\rho g = 7.85\times10^{-5}$ N/mm^3 acting in $-y$. $tA = 5\times600 = 3000$ mm^3, so total weight $W = 0.2355$ N. Nodal load $f_y = -tA\,\rho g/3 = -0.0785$ N at each node (x-components zero).

(b) **Pressure** $p = 10$ N/mm^2 acting normal to the inclined edge 2-3 (length 50 mm), pointing inwards. Direction cosines of the outward normal : (0.6, 0.8) $\Rightarrow$ $T_x = -6$, $T_y = -8$ N/mm^2. $f = tL_e/2\,[T_x, T_y] = (5\times50/2) = 125 \times [-6, -8] = [-750, -1000]$ N at each of nodes 2 and 3; node 1 gets zero.

$$\{f_T\} = \begin{bmatrix} 0 \\ 0 \\ -750 \\ -1000 \\ -750 \\ -1000 \end{bmatrix}$$

Check : resultant $= \sqrt{(1500^2 + 2000^2)} = 2500$ N $= p\,t\,L = 10 \times 5 \times 50$ OK (acting on the edge, normal to it).

Problem 5 : plane strain [D] and the effect of ν

For $E = 2\times10^5$ N/mm^2, $\nu = 0.3$:

$$\text{Plane stress } [D]\ (\times10^5) = \begin{bmatrix} 2.198 & 0.659 & 0.000 \\ 0.659 & 2.198 & 0.000 \\ 0.000 & 0.000 & 0.769 \end{bmatrix}$$

$$\text{Plane strain } [D]\ (\times10^5) = \begin{bmatrix} 2.692 & 1.154 & 0.000 \\ 1.154 & 2.692 & 0.000 \\ 0.000 & 0.000 & 0.769 \end{bmatrix}$$

Plane strain is stiffer ($D_{11} = 2.692\times10^5$ against 2.198×10^5). In the plane-strain case $\sigma_z = \nu(\sigma_x + \sigma_y) \neq 0$ must be added when finding von Mises stress.

Problem 6 : thermal load on a CST

Plane stress triangle, $t = 5$, $A = 600\ mm^2$, $\alpha = 12\times10^{-6}/°C$, free expansion $\Delta T = 100\ °C$ produces initial strain $\varepsilon_0 = [1.2\times10^{-3},\ 1.2\times10^{-3},\ 0]^T$. Thermal load $\{f_{th}\} = tA[B]^T[D]\{\varepsilon_0\}$. $[D]\{\varepsilon_0\} = (E/(1-\nu))\ \alpha\Delta T\ [1\ 1\ 0]^T = 2.857\times10^5\times1.2\times10^{-3} = 342.9\ N/mm^2$ in both σ_x, σ_y. Then $f = tA/(2A)\ [b_1, c_1, b_2, c_2, b_3, c_3]\times342.9 = (t/2)(342.9)[b_1\ c_1\ b_2\ c_2\ b_3\ c_3]^T$ (b, c in mm). If the element is completely free, the displacements absorb the load and $\sigma = 0$.

Problem 7 : axisymmetric loads on a thick cylinder

Thick cylinder, mean section : bore pressure $p = 10\ N/mm^2$ acting on the inner surface $r = 50$ mm over an edge of length 20 mm (nodes i, j both at $r = 50$) :

$$f_i = f_j = \pi\ p\ L_e(2r_i + r_j)/3 = \pi\times10\times20\times150/3 = 31416\ N\ (radial)$$

Check with $2\pi\ r\ p\ L_e/2 = \pi\times50\times10\times20 = 31416\ N$ OK. For a spinning disc ($\rho = 7.85\times10^{-9}\ N\cdot s^2/mm^4$, $\omega = 300\ rad/s$) the body force at the element of Problem 3 is $F_r = \rho\omega^2\bar{r} = 7.85\times10^{-9}\times9\times10^4\times23.33 = 0.0165\ N/mm^3$; total radial force = $F_r \times 2\pi\ \bar{r}\ A = 120.8\ N$, i.e. $40.3\ N$ at each node.

Higher-order triangle (LST) in brief

- 6-noded triangle (3 corner + 3 mid-side nodes) : complete quadratic $u = \alpha_1 + \alpha_2x + \alpha_3y + \alpha_4x^2 + \alpha_5xy + \alpha_6y^2$, 12 dof.
- Strain varies **linearly** (Linear Strain Triangle) so it models bending and stress concentrations much better than the CST; shape functions in area co-ordinates : $N_1 = L_1(2L_1 - 1)$, $N_4 = 4L_1L_2$, ...

Short answer questions

Q1 Why is CST called constant strain triangle ? - $[B]$ contains only nodal coordinates, so ε and σ are constant over the element.

Q2 Size of $[k]$ for CST ? - 6×6 ; for a 4-noded quadrilateral 8×8.

Q3 Why must nodes be numbered anticlockwise ? - To obtain positive area A ($2A$ is the determinant).

Q4 Why does $[B]$ of the axisymmetric element depend on r, z ? - Hoop strain $\varepsilon_\theta = u/r$.

Q5 What is the volume element of an axisymmetric solid ? - $dV = 2\pi\ r\ dr\ dz$.

Q6 Where do we use plane strain ? - Dams, retaining walls, long pipes, tunnels (ε_z

$= 0$).

Problem 8 : stress in an axisymmetric ring element

For the triangle of Problem 3, let the nodal displacements be $u_1 = 0.004$, $w_1 = 0$, $u_2 = 0.006$, $w_2 = 0$, $u_3 = 0.004$, $w_3 = 0.002$ mm. Then $\{\varepsilon\} = [B]\{q\}$ gives

ε_r	ε_z	ε_θ	γ_{rz}
0.000200	0.000200	0.000200	0.000000

and $\{\sigma\} = [D]\{\varepsilon\}$: $\sigma_r = 100.0$, $\sigma_z = 100.0$, $\sigma_\theta = 100.0$, $\tau_{rz} = 0.0$ N/mm^2. Note that $\sigma_\theta \neq 0$ (hoop stress) even though no circumferential displacement exists.

Problem 9 : equivalent load of a linearly varying edge pressure

Pressure varies from p_i at node i to p_j at node j along an edge of length L_e (thickness t). Using the linear shape functions along the edge :

$$f_i = t\,L_e(2p_i + p_j)/6 \quad f_j = t\,L_e(p_i + 2p_j)/6$$

Example : $t = 5$, $L_e = 30$, $p_i = 0$, $p_j = 12$ N/mm^2 : $f_i = 5\times30\times12/6 = 300$ N, $f_j = 5\times30\times24/6 = 600$ N; total 900 N $= t\,L_e(p_i+p_j)/2$. Always check that the total of nodal loads equals the resultant of the distributed load.

Complete procedure for a 2-D (plane) problem

1. Idealise : decide plane stress / plane strain / axisymmetric; use symmetry to cut the model.
2. Mesh : number nodes and elements (anticlockwise); tabulate connectivity and coordinates.
3. For each element compute b_i, c_i, A, $[B]$, $[D]$, $[k] = tA\,B^TDB$.
4. Element load vectors (body force, traction, thermal) and point loads.
5. Assemble $[K]$ (dof of node $i = 2i{-}1,\ 2i$) and $\{F\}$.
6. Apply BC, solve for $\{Q\}$.
7. Recover $\{\varepsilon\} = [B]\{q\}$, $\{\sigma\} = [D]\{\varepsilon\}$, principal and von Mises stress, reactions.

Element comparison

Element	Nodes	dof	Polynomial	Strain
CST	3	6	linear	constant
LST	6	12	quadratic	linear
Q4	4	8	bilinear (1,x,y,xy)	linear in part
Q8	8	16	serendipity quad.	quadratic
Axisym. tri.	3	6	linear in r,z	varies (use centroid)

Tips: Boundary-condition tips : symmetry plane ⇒ normal displacement = 0 ; antisymmetry ⇒ tangential displacement = 0 ; for axisymmetric solids, nodes on the axis have $u = 0$. Never leave a rigid-body mode (3 in plane, 4 in 3-D axisymmetric with axial translation).

Mesh guidelines and expected questions

- Use fine mesh where stress gradients are high (holes, notches, fillets, load points); avoid very thin triangles (aspect ratio < 3). Make node on every point load, material change, support.
- Derive shape functions, [B] and [k] of a CST; explain plane stress vs plane strain with [D]. Solve numerical problems (1-2 elements) for stresses, load vector.
- Derive [B], [D] and [k] for a triangular axisymmetric element; why is [B] evaluated at the centroid ? Give load vector for pressure and centrifugal force.